EC 704 - Macroeconomic Theory 2

Problem Set 1

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Problem 1

Part 1

The continuous time limit of the unemployment rate evolution is

$$\dot{u}(t) = -f_t u(t) + s_t(1 - u(t)) \tag{1}$$

Part 2

Rewriting above equation for $t \in [m, m+1)$,

$$\begin{aligned}
\dot{u}(t) + (f_m + s_m)u(t) &= s_m \\
\implies \frac{d}{dt}u(t)e^{(f_m+s_m)t} &= s_m e^{(f_m+s_m)t} \\
\implies u(t) &= \frac{s_m}{f_m + s_m} + Ce^{-(f_m+s_m)t}
\end{aligned}$$

setting $u(t_m) = u_m$ gives

$$u(t) = \frac{s_m}{f_m + s_m} + \left(u_m - \frac{s_m}{f_m + s_m}\right) e^{-(f_m+s_m)(t-t_m)} \tag{2}$$

which gives

$$\begin{aligned}
u_{m+1} &= \frac{s_m}{f_m + s_m} + \left(u_m - \frac{s_m}{f_m + s_m}\right) e^{-(f_m+s_m)} \\
\implies u_{m+1} - u_m &= -\underbrace{\frac{f_m}{f_m + s_m}\left(1 - e^{-(f_m+s_m)}\right)}_{=F_m} u_m + \underbrace{\frac{s_m}{f_m + s_m}\left(1 - e^{-(f_m+s_m)}\right)}_{=S_m}(1 - u_m)
\end{aligned}$$

Which gives expressions

$$F_m = \frac{f_m}{f_m + s_m}\left(1 - e^{-(f_m+s_m)}\right) \tag{3}$$

$$S_m = \frac{s_m}{f_m + s_m}\left(1 - e^{-(f_m+s_m)}\right) \tag{4}$$

Part 3

The time aggregation bias are

$$\begin{aligned}
F_m - f_m &= -f_m\left(1 - \frac{1}{f_m + s_m}\left(1 - e^{-(f_m+s_m)}\right)\right) \\
S_m - s_m &= -s_m\left(1 - \frac{1}{f_m + s_m}\left(1 - e^{-(f_m+s_m)}\right)\right)
\end{aligned}$$

These expressions show that there is a negative bias in the monthly measured job-finding and job-losing rates. This is precisely because it is possible that workers find a job and then lose it within a month. When both f_m and s_m are small, the bias goes to zero as $\frac{1-e^{-x}}{x} \approx 1$. This is because the probability of getting a job and then losing it within the same month is of second order.

If either f_m or s_m is large, then both job-finding and job-losing rates have equal relative bias.

Problem 2

I am studying the heterogeneity on education using the categories - High school dropout, High school graduate, Some college, College completed (4+ years), Post graduate.

Part 1

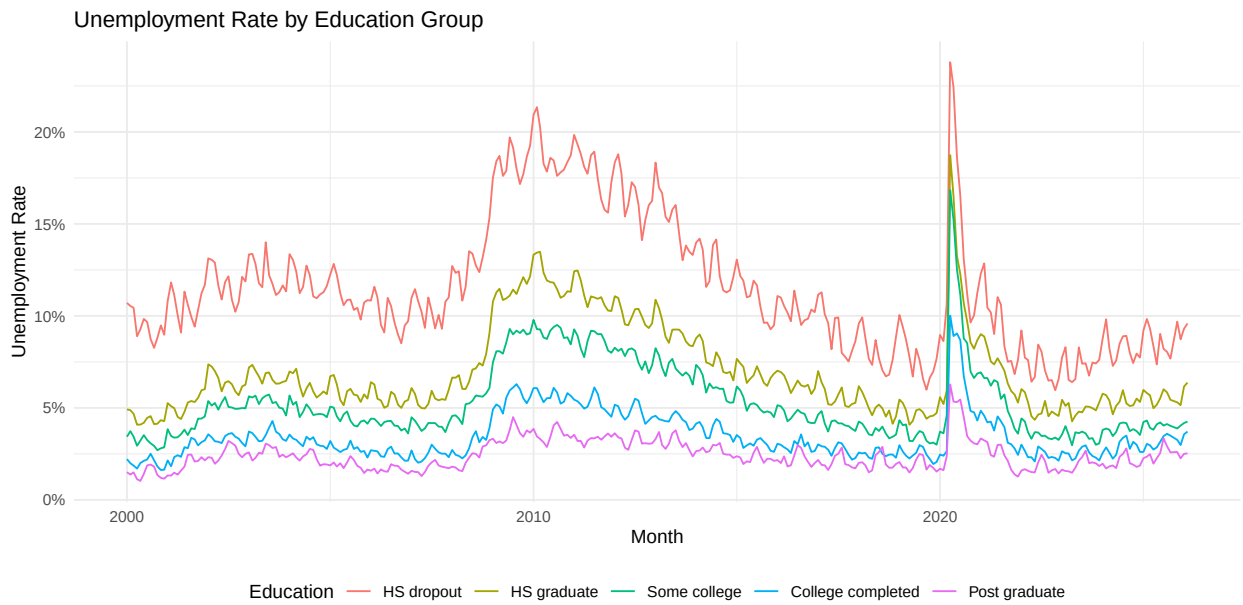

Figure 1: Unemployment rate by education group

The figure shows that unemployment rates are consistently lower for more educated groups.

Table 1: Average Unemployment Rate by Education Group

Education Group	**Avg. Unemployment Rate**	**Avg.** N
HS dropout	0.116	6098
HS graduate	0.070	17943
Some college	0.053	17916
College completed	0.034	13607
Post graduate	0.024	7512

Average unemployment rate for all participants in labor force over the period is 5.65%.

Table 2: Changes in Unemployment Rates by Education Group during Great Recession

Education Group	u_0	u_T	Δu	N_0	N_T
HS dropout	0.110	0.197	0.087	7221	7721
HS graduate	0.060	0.111	0.051	20462	20556
Some college	0.042	0.093	0.051	20060	20645
College completed	0.023	0.059	0.036	13895	14360
Post graduate	0.017	0.038	0.021	7300	7541

Table 3: Changes in Unemployment Rates During COVID-19 by Education Group

Education Group	u_0	u_T	Δu	N_0	N_T
HS dropout	0.086	0.238	0.152	4311	3297
HS graduate	0.052	0.187	0.135	15355	12020
Some college	0.036	0.168	0.132	16369	13371
College completed	0.024	0.100	0.076	14467	12169
Post graduate	0.016	0.063	0.047	8480	7333

Part 2

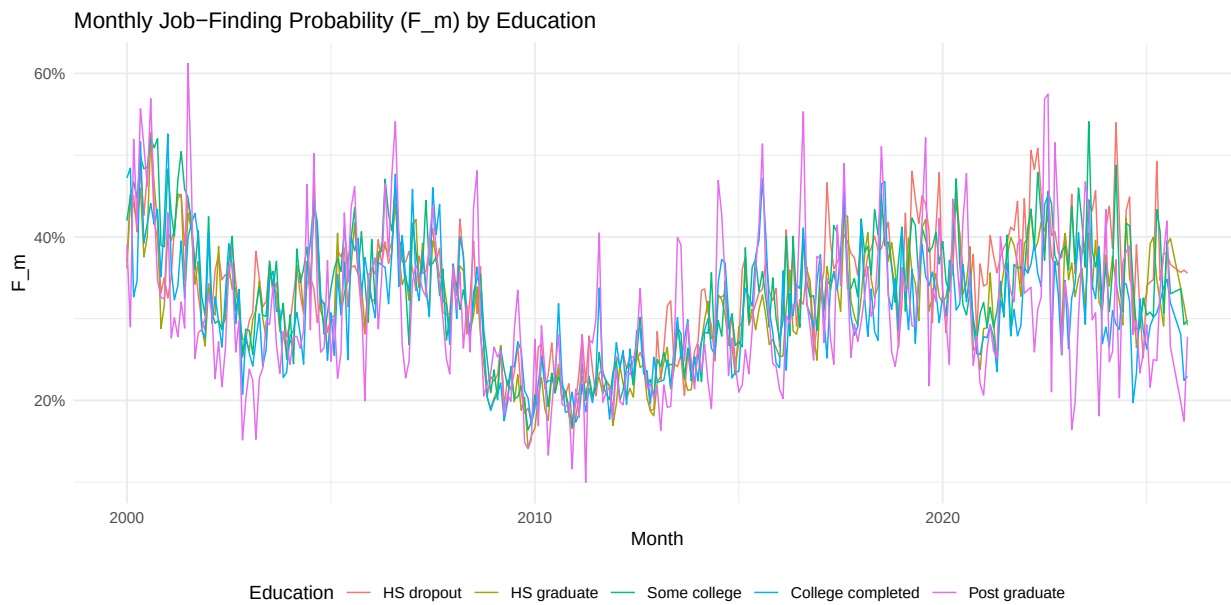

Figure 2: Monthly job finding probability F_m by education criteria

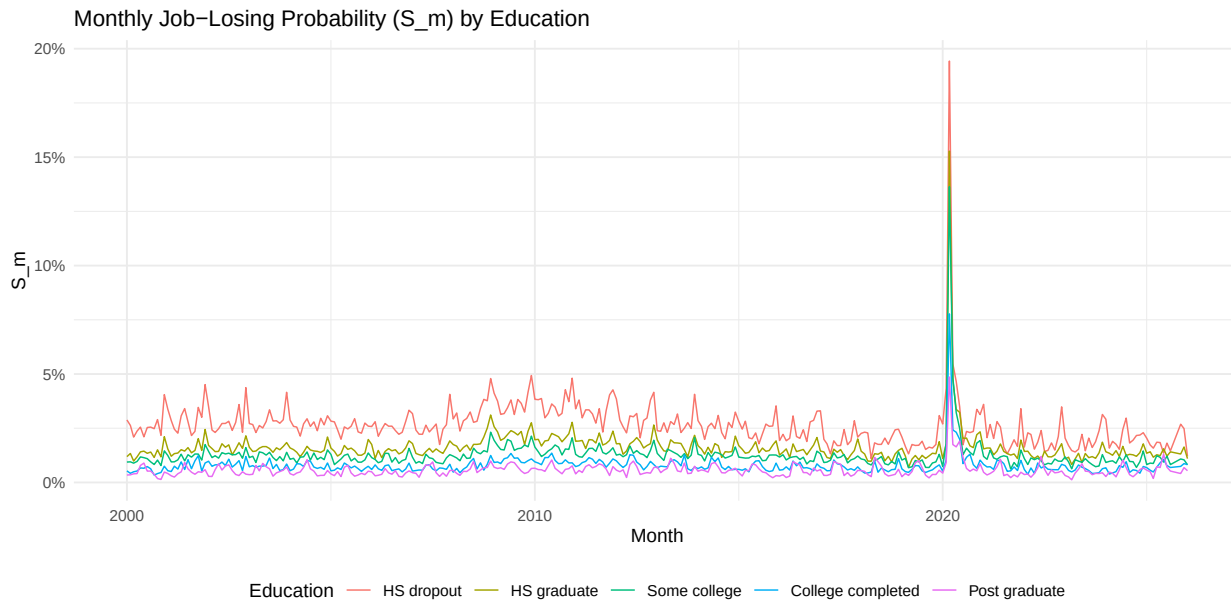

Figure 3: Monthly job losing probability S_m by education criteria

Part 3

From equations 3 and 4

$$\frac{F_m}{S_m} = \frac{f_m}{s_m}$$
$$F_m + S_m = 1 - e^{-(f_m + s_m)}$$

solving these, we get

$$f_m = -\ln(1 - F_m - S_m)\frac{F_m}{F_m + S_m}$$
$$s_m = -\ln(1 - F_m - S_m)\frac{S_m}{F_m + S_m}$$

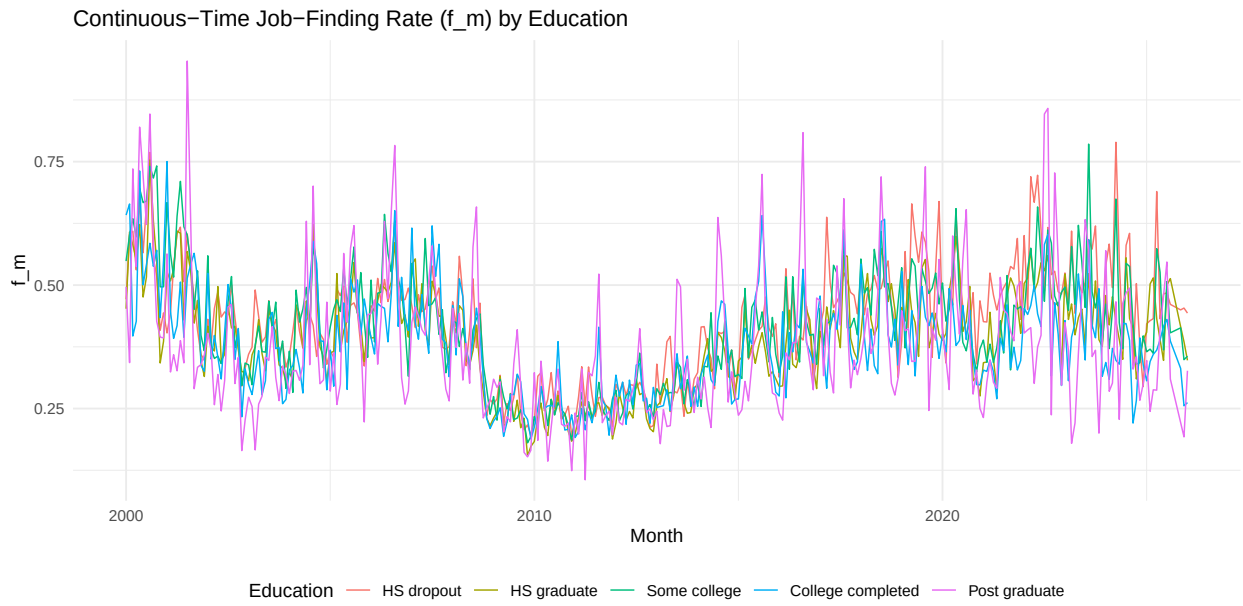

Figure 4: Underlying job finding rates f_m by education criteria

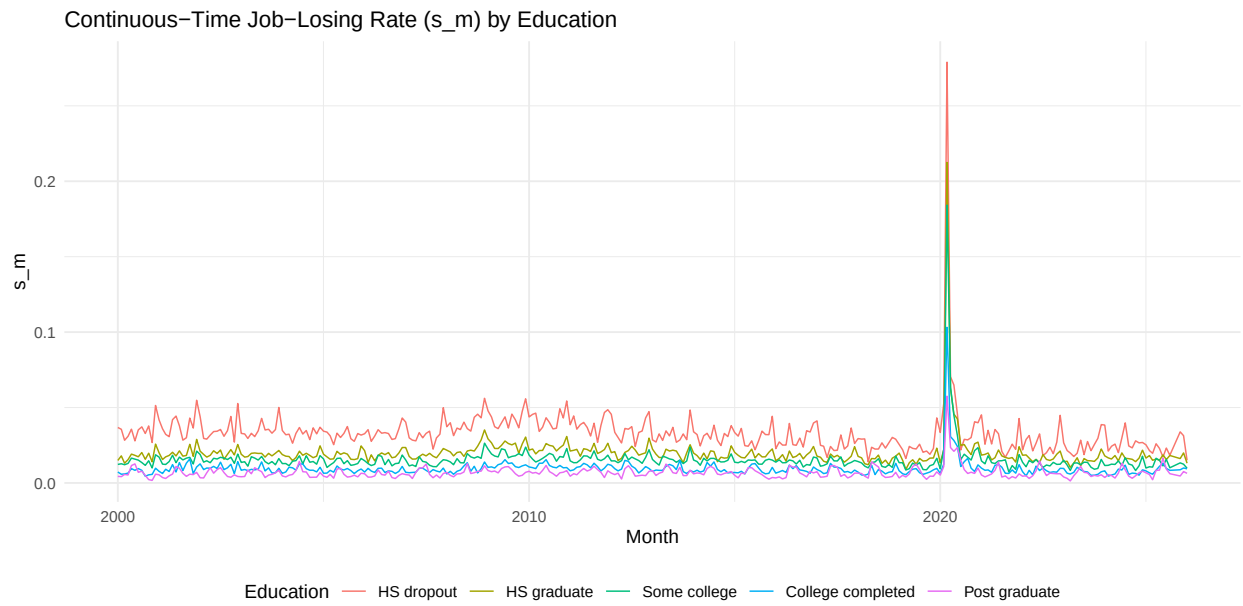

Figure 5: Underlying job losing rates s_m by education criteria

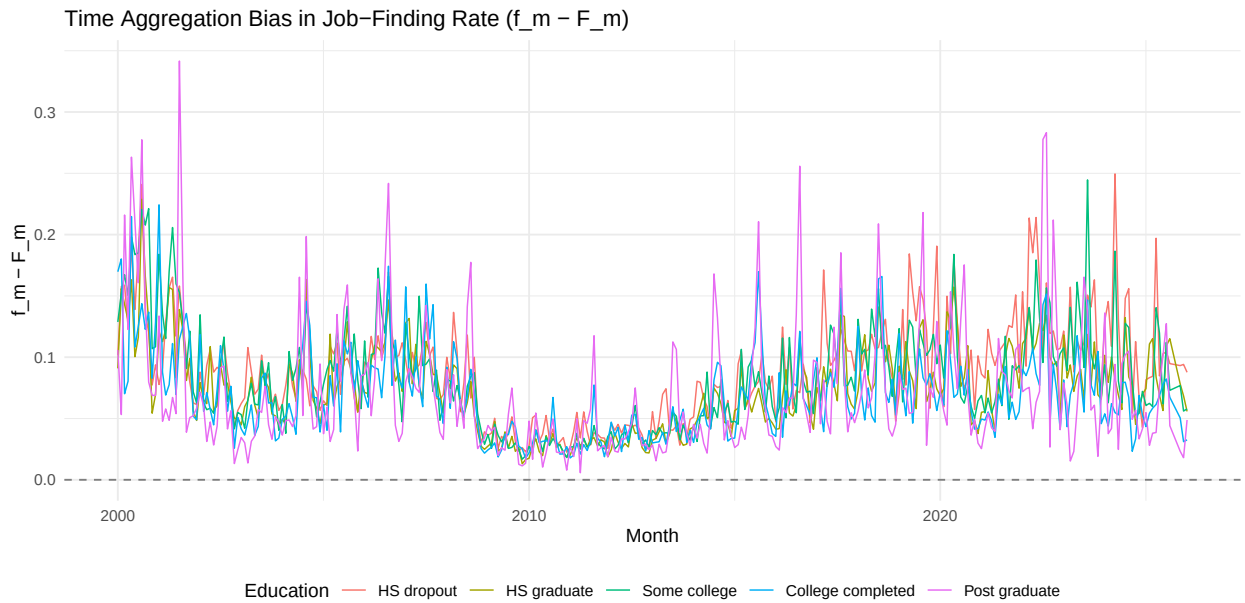

Figure 6: Aggregation bias in job finding rates $f_m - F_m$ by education criteria

Figure 7: Aggregation bias in job losing rates $s_m - S_m$ by education criteria

Part 4

In month m, following lecture notes, define the quasi-equilibrium unemployment rate of education group g as

$$\hat{u}_m^g = \frac{S_m^g}{S_m^g + F_m^g} \tag{5}$$

The share of fluctuation is given by

$$\text{flutuations due to S in month m} = \frac{Cov_g\left[\log \frac{\hat{u}_m^g}{1-\hat{u}_m^g}, \log s_m^g\right]}{Var_g\left[\log \frac{\hat{u}_m^g}{1-\hat{u}_m^g}\right]}$$

$$\text{flutuations due to F in month m} = \frac{Cov_g\left[\log \frac{\hat{u}_t^g}{1-\hat{u}_t^g}, -\log f_m^g\right]}{Var_g\left[\log \frac{\hat{u}_m^g}{1-\hat{u}_m^g}\right]}$$

where Cov_g and Var_g represent cross sectional covariance and variance weighted by group weights. I use S_m, F_m (discrete-time probabilities) in place of s, f.

Levels decomposition

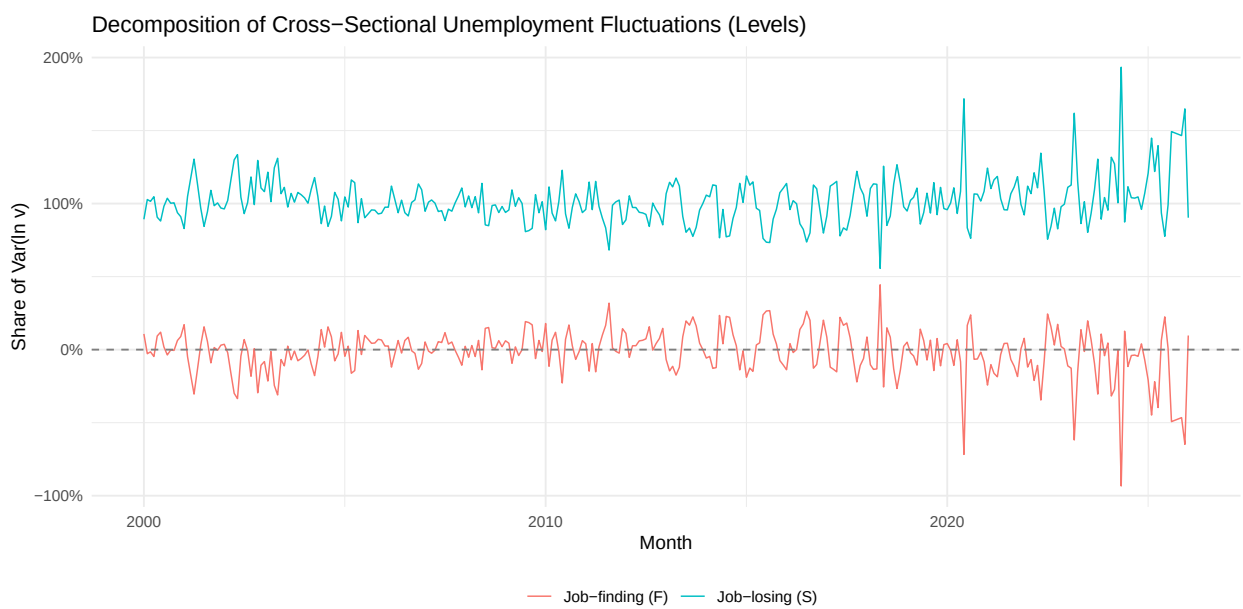

Figure 8: Decomposition of cross-sectional unemployment fluctuations (levels)

Table 4: Mean Share of Fluctuations (Levels)

Component	**Mean Share**
Job-losing (S)	101.9%
Job-finding (F)	-1.9%
Sum	100%

In the levels decomposition, job-losing accounts for slightly more than 100% of the cross-sectional variance in steady-state unemployment across education groups, while job-finding has a small negative share (-1.9%). This implies that across education groups, differences in S more than fully explain unemployment dispersion, and differences in F actually slightly compress it — groups with higher S also tend to have higher F, partially offsetting the effect of job-losing on unemployment.

Changes decomposition

I also perform the same decomposition on monthly changes $\Delta \ln v$, $\Delta \ln S$, $\Delta \ln F$.

$$\text{flutuations due to S in month m} = \frac{Cov_g\left[\Delta \log \frac{\hat{u}_m^g}{1-\hat{u}_m^g}, \Delta \log s_m^g\right]}{Var_g\left[\Delta \log \frac{\hat{u}_m^g}{1-\hat{u}_m^g}\right]}$$

$$\text{flutuations due to F in month m} = \frac{Cov_g\left[\Delta \log \frac{\hat{u}_t^g}{1-\hat{u}_t^g}, -\Delta \log f_m^g\right]}{Var_g\left[\Delta \log \frac{\hat{u}_m^g}{1-\hat{u}_m^g}\right]}$$

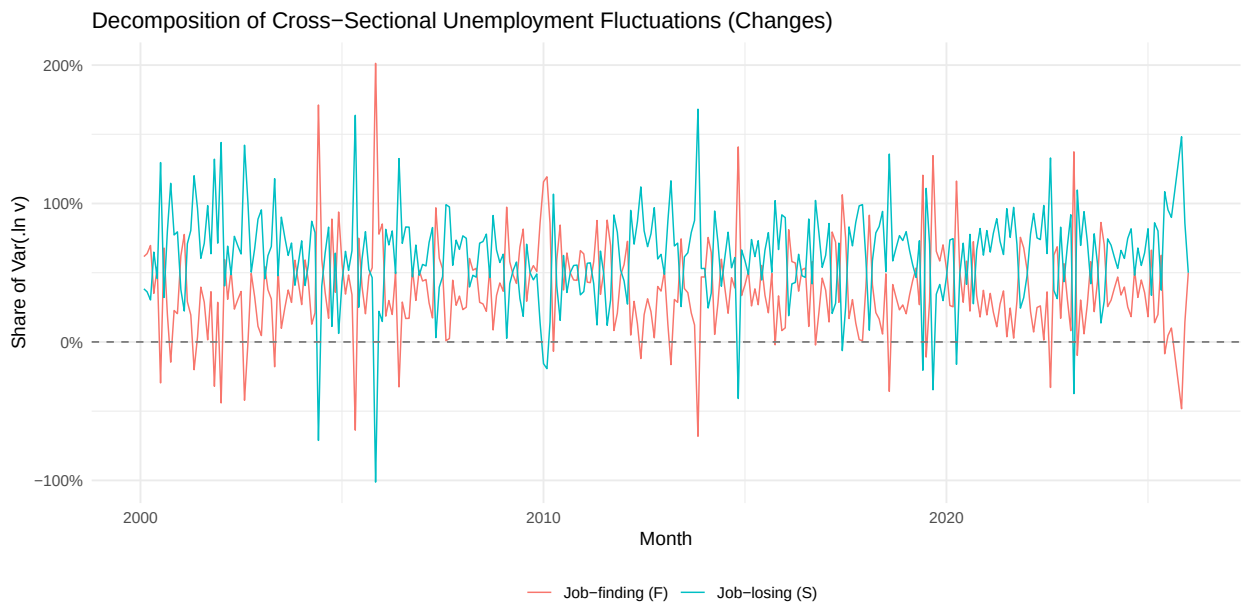

Figure 9: Decomposition of cross-sectional unemployment fluctuations (monthly changes)

Table 5: Mean Share of Fluctuations (Monthly Changes)

Component	**Mean Share**
Job-losing (S)	62.0%
Job-finding (F)	38.0%
Sum	100 %

In the changes decomposition, job-losing accounts for about 62% of the cross-sectional variance in monthly changes in steady-state unemployment, while job-finding accounts for 38%. Both components contribute positively, indicating that month-to-month changes in the cross-sectional dispersion of unemployment are driven by both margins, with job-losing being the dominant source.

Problem 3

Part 1

$$u = 0.05 \implies f(\theta) = s\frac{1-u}{u} = 0.38$$

$$f(\theta) = M(1,\theta) = \frac{\theta}{(1+\theta^{\iota})^{1/\iota}}$$
$$\implies \theta = 0.4414$$

Hence $\theta = 0.4414$ is consistent with $u = 0.05$.

From the free entry and wage condition in steady state

$$c = \beta q(\theta)\frac{1}{1-\beta(1-s)}[(1-\gamma)(z-b) - \gamma\theta c]$$
$$\implies c = 1.211$$

Part 2

Transition matrix is given by

$$P = \begin{bmatrix} \pi & ,1-\pi \\ 1-\pi & ,\pi \end{bmatrix}$$

It has a unique stationary state

$$p = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$

Since $\mathbb{E}[\log z] = 0$ and $Var[\log z] = 0.0001$, we have

$$\log z_h = -\log z_l$$
$$(\log z_h)^2 + (\log z_l)^2 = 2 * 0.0001$$

which gives

$$z_h = e^{0.01} \quad , \quad z_l = e^{-0.01}$$

The autocorrelation is

$$\rho = \frac{\mathbb{E}[\log z_t \log z_{t+1}]}{\text{Var}(\log z_t)}$$
$$= \frac{\mathbb{E}\left[\log z_t \mathbb{E}[\log z_{t+1} \mid \log z_t]\right]}{\text{Var}(\log z_t)}$$

Substituting $\mathbb{E}[\log z_{t+1} \mid \log z_t] = \pi \log z_t - (1-\pi)\log z_t$ gives

$$\rho = 2\pi - 1 \implies \pi = 0.985$$

Part 3

Free entry condition and value of filling a position is

$$c = \beta q(\theta(z,u))\mathbb{E}_{z',u'}[J(z',u')|z]$$
$$J(z,u) = z - w(z,u) + \beta(1-s)\mathbb{E}_{z',u'}[J(z',u')|z]$$

where $w(z,u) = (1-\gamma)b + \gamma(z + \theta(z,u)c)$.

Substituting the formul for $J(z,u)$ and $w(z,u)$ in first equation gives two relationships between c, z_h, z_l, $\theta(z_h,u)$ and $\theta(z_l,u)$. For fixed c, since these equations do not directly depend on u, θ is a function of z only. Hence, w also does not directly depend on u, and $J(z,u)$ is a function of z only. We can write $\theta(z)$ and $J(z)$.
We have

$$q(\theta) = M(1/\theta, 1) = \frac{1}{(1+\theta^\iota)^{1/\iota}}$$

The 4 equations in terms of $\{\theta(z_l), \theta(z_h), J(z_l), J(z_h)\}$ are

$$\frac{c}{\beta}(1+\theta(z_h)^\iota)^{\frac{1}{\iota}} = \pi J(z_h) + (1-\pi)J(z_l)$$
$$\frac{c}{\beta}(1+\theta(z_l)^\iota)^{\frac{1}{\iota}} = \pi J(z_l) + (1-\pi)J(z_h)$$
$$J(z_h) = (1-\gamma)(z_h - b) - \gamma c\theta(z_h) + \beta(1-s)[\pi J(z_h) + (1-\pi)J(z_l)]$$
$$J(z_l) = (1-\gamma)(z_l - b) - \gamma c\theta(z_l) + \beta(1-s)[\pi J(z_l) + (1-\pi)J(z_h)]$$

Solving these gives

$$\theta(z_h) = 0.448925$$
$$\theta(z_l) = 0.433358$$
$$J(z_h) = 1.416369$$
$$J(z_l) = 1.405458$$

The simulation results are in figure 10. The standard deviations are

$$sd(\log z_t) = 0.010$$
$$sd(\log u_t) = 0.013$$
$$sd(\log v_t) = 0.007$$
$$sd(\log \theta_t) = 0.018$$

Assuming perfect correlation, $|d\log u| \approx 1.3|d\log u|$, which is much smaller than the value overserved in data.

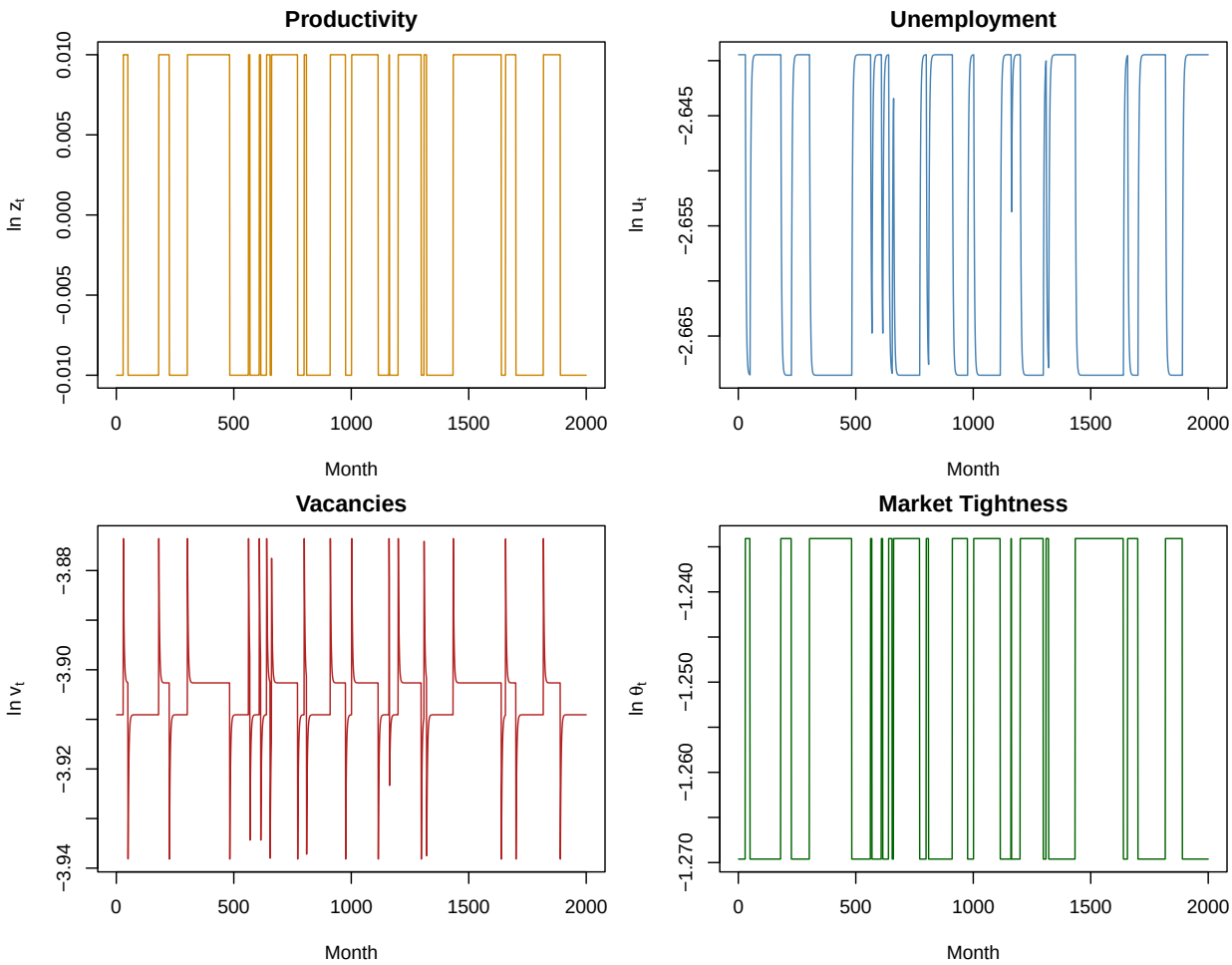

Figure 10: Simulation Results for $b = 0.4$

Part 4

With $b = 0.95$, we have the new params

$$
\begin{aligned}
c &= 0.101 \\
\theta(z_h) &= 0.534409 \\
\theta(z_l) &= 0.347724 \\
J(z_h) &= 0.123372 \\
J(z_l) &= 0.112493
\end{aligned}
$$

the simulation results are in figure 11. The standard deviations are

$$sd(\log z_t) = 0.010$$
$$sd(\log u_t) = 0.156$$
$$sd(\log v_t) = 0.083$$
$$sd(\log \theta_t) = 0.215$$

This restores $|d \log u| \approx 15.6|d \log u|$, which is more closer to the value observed in the data. Therefore, considering outside option to be larger and closer to worker productivity seem to resolve the Shimer puzzle.

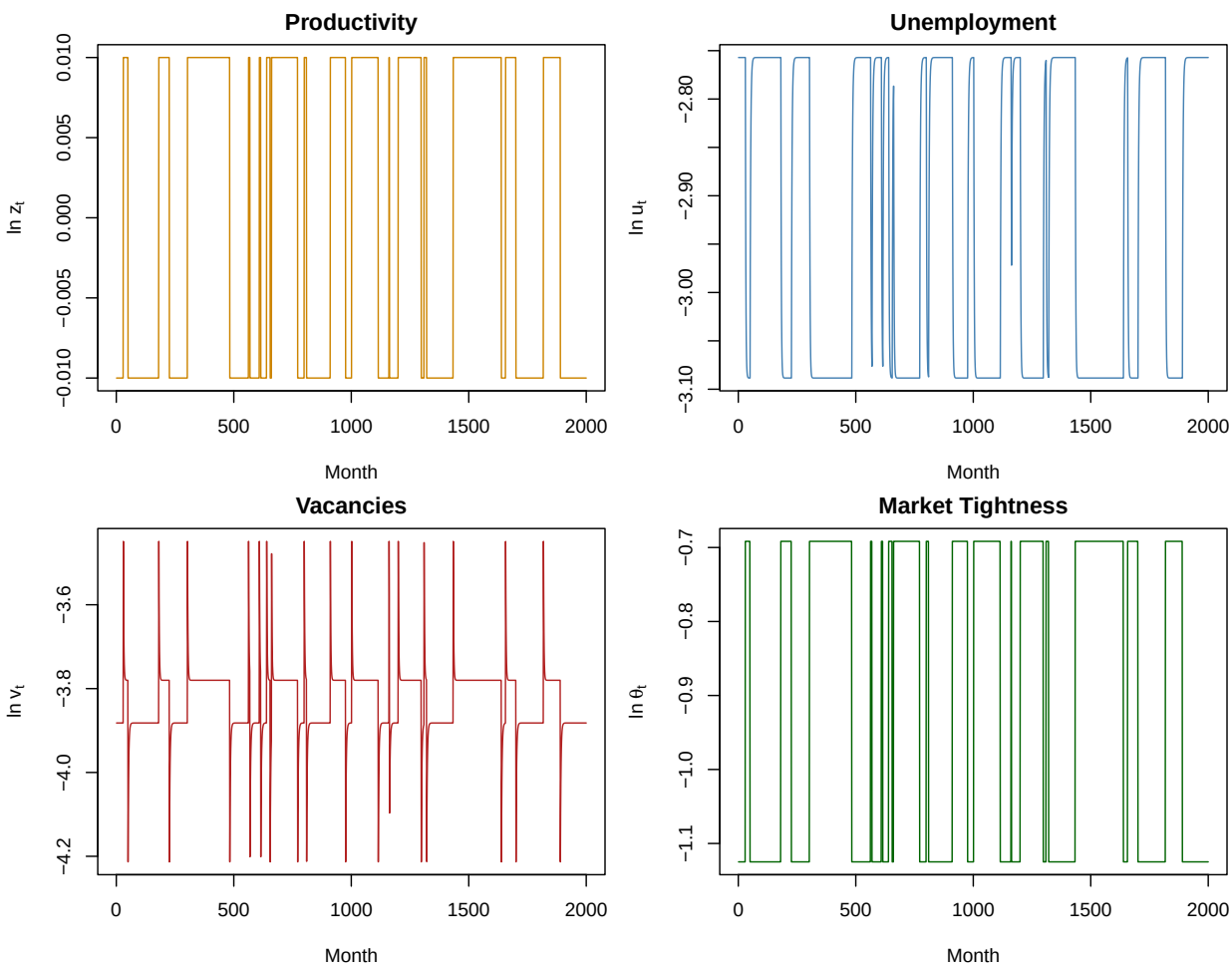

Figure 11: Simulation Results for $b = 0.95$

Part 5

Let flow utility be $b = \bar{b} z_t$. Then the 4 equations are

$$\frac{c}{\beta}(1+\theta(z_h)^{\iota})^{\frac{1}{\iota}} = \pi J(z_h) + (1-\pi)J(z_l)$$
$$\frac{c}{\beta}(1+\theta(z_l)^{\iota})^{\frac{1}{\iota}} = \pi J(z_l) + (1-\pi)J(z_h)$$
$$J(z_h) = (1-\gamma)(1-\bar{b})z_h - \gamma c\theta(z_h) + \beta(1-s)[\pi J(z_h) + (1-\pi)J(z_l)]$$
$$J(z_l) = (1-\gamma)(1-\bar{b})z_l - \gamma c\theta(z_l) + \beta(1-s)[\pi J(z_l) + (1-\pi)J(z_h)]$$

which gives the solutions

$$c = 0.101$$
$$\theta(z_h) = 0.445394$$
$$\theta(z_l) = 0.436061$$

the simulation results are in figure 12. The standard deviations are

$$sd(\log z_t) = 0.010$$
$$sd(\log u_t) = 0.008$$
$$sd(\log v_t) = 0.004$$
$$sd(\log \theta_t) = 0.011$$

Now we only have $|d\log u| \approx 0.8|d\log u|$, which worsens the Shimer puzzle. If the outside option is worse during a downturn, then worker becomes more likely to work for lower wage which counters the unemployment effect of low productivity.

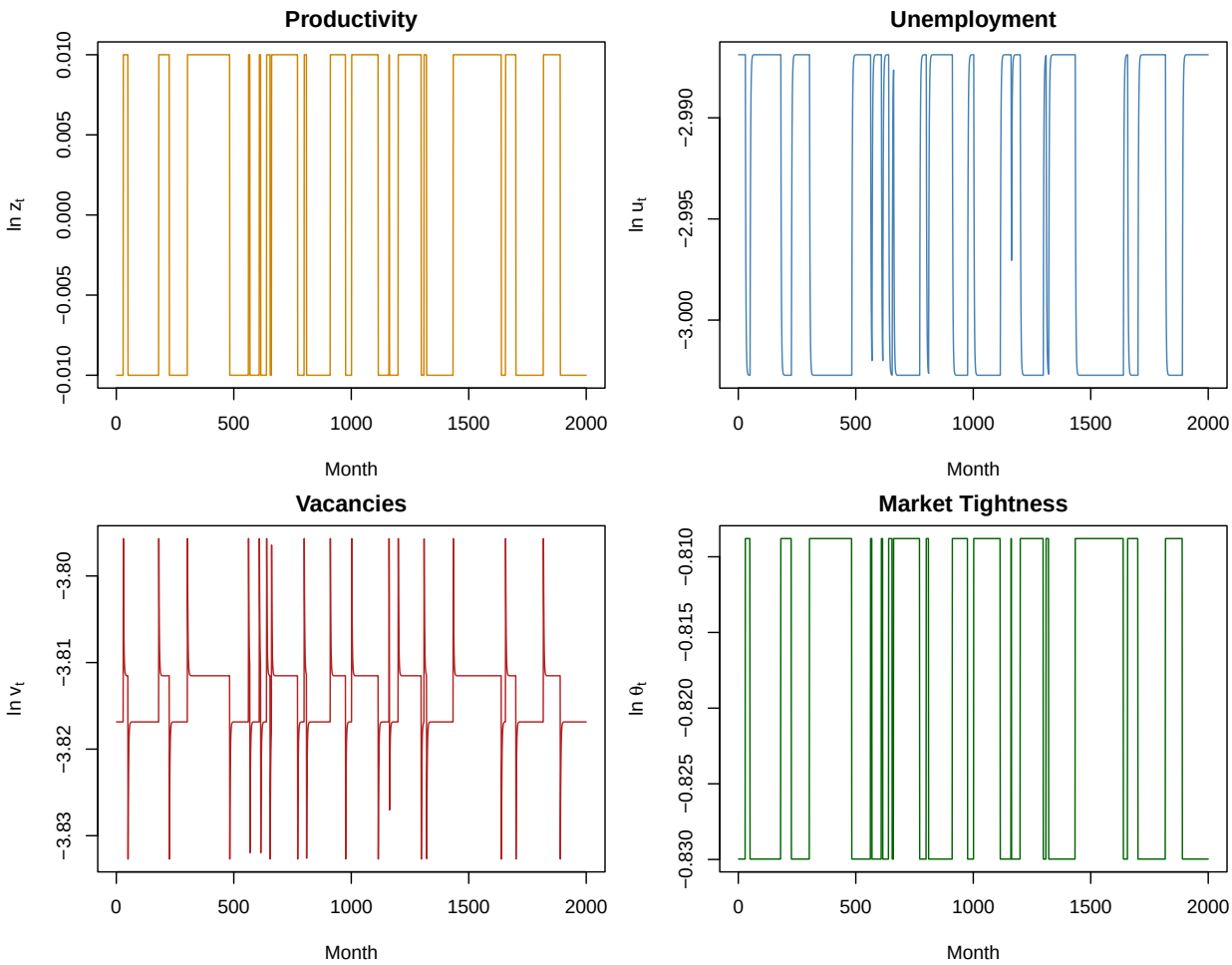

Figure 12: Simulation Results for $b_t = 0.95 \times z_t$

Part 6

Let rigid wages be $w_t = \bar{w} = 0.95$. Then the 4 equations are

$$\frac{c}{\beta}(1+\theta(z_h)^{\iota})^{\frac{1}{\iota}} = \pi J(z_h) + (1-\pi)J(z_l)$$
$$\frac{c}{\beta}(1+\theta(z_l)^{\iota})^{\frac{1}{\iota}} = \pi J(z_l) + (1-\pi)J(z_h)$$
$$J(z_h) = z_h - \bar{w} + \beta(1-s)[\pi J(z_h) + (1-\pi)J(z_l)]$$
$$J(z_l) = z_l - \bar{w} + \beta(1-s)[\pi J(z_l) + (1-\pi)J(z_h)]$$

Also, fixed wages, formula for c in steady state changes and can be computed by equations (3-4) and (3-5)

$$J(z) = \frac{z - \bar{w}}{1 - \beta(1-s)}$$
$$c = \beta q(\theta) \times \frac{z - \bar{w}}{1 - \beta(1-s)}$$

Using $\theta = 0.4414$ as before gives $c = 1.887$
which gives the solutions

$$c = 1.840$$
$$\theta(z_h) = 0.607849$$
$$\theta(z_l) = 0.239225$$

the simulation results are in figure 13. The standard deviations are

$$sd(\log z_t) = 0.010$$
$$sd(\log u_t) = 0.344$$
$$sd(\log v_t) = 0.180$$
$$sd(\log \theta_t) = 0.466$$

This shows that completely rigid wages solve the Shimer puzzle, bringing $|d \log u| \approx 0.8|d \log u|$. With rigid wages, firm's value function of a filled position fluctuates a lot with worker productivity, which results in more firms closing down during recession and more vacancies being created in boom.

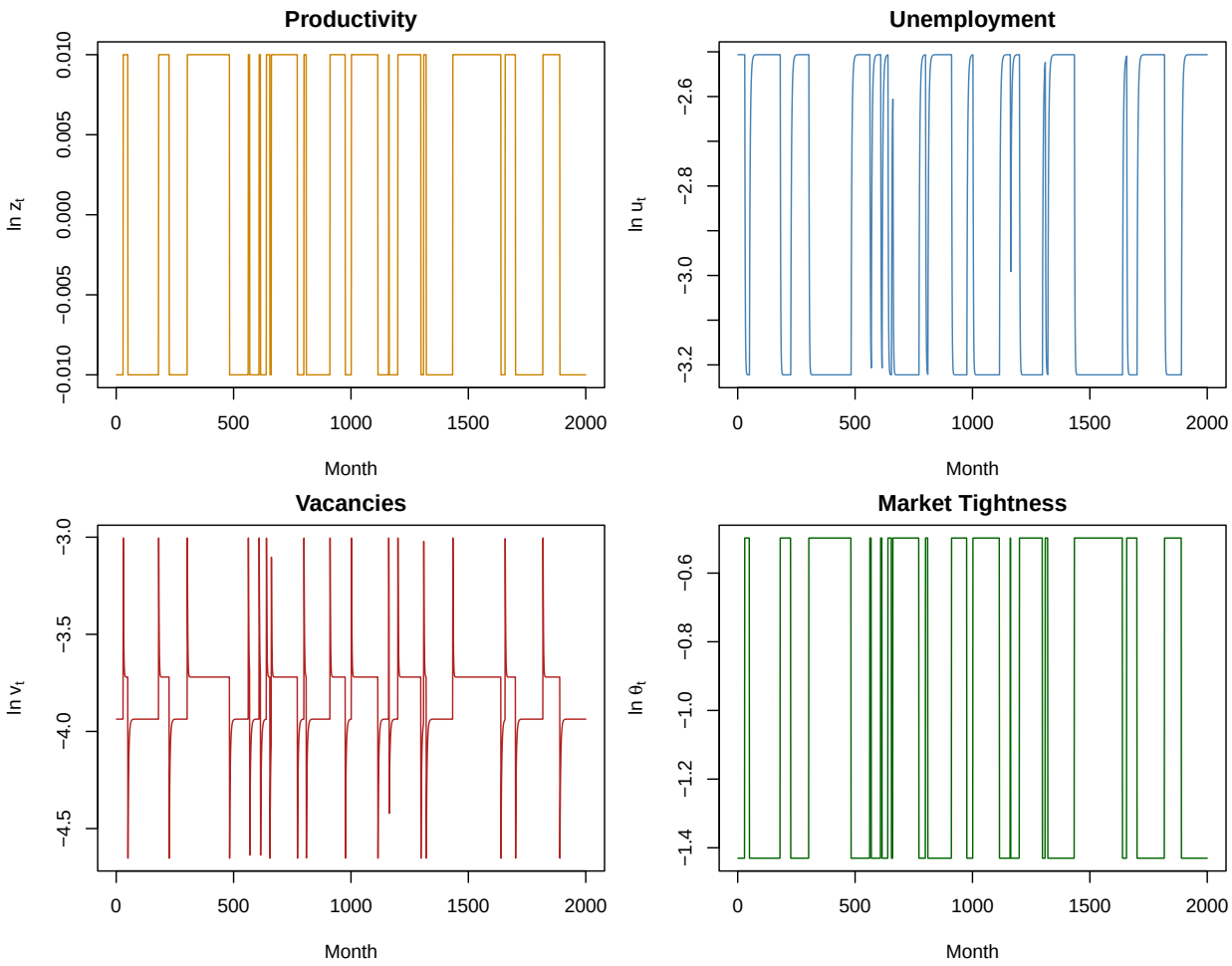

Figure 13: Simulation Results for $w_t = \bar{w} = 0.95$